

APPENDIX L

Magnetic-hysteresis damping

This chapter discusses magnetic-hysteresis damping. This is an approach often used in lower-cost satellites. When used with a magnetic torque rod they can dissipate energy without any moving parts and without any additional power, aside from the power for the torque rod.

L.1. Magnetic-hysteresis damper model

Hysteresis dampers are passive and act continuously to damp angular rates. The flux vector through the rod, $\mathbf{B}$, can be used to calculate the rod's magnetic dipole, $\mathbf{M}_D$, from

$$\mathbf{M}_D = \mathbf{B}V \quad (\text{L.1})$$

where V is volume of the hysteresis rods. The flux is a function of the applied magnetic field $\mathbf{H}$, the vacuum permeability $\mu_0 = 4\pi \times 10^{-7}$ H/m, and the apparent permeability of the rod μ'_r . μ'_r is crucial in determining the effect of finite geometric constraints on the magnetic properties; it is a function of the true material permeability μ_r and the demagnetizing factor, N_d ,

$$\mathbf{B} = \mu_0 \mu'_r \mathbf{H} \quad (\text{L.2})$$

$$\mu'_r = \frac{\mu_r(\mathbf{H})}{1 + N_d \mu_r(\mathbf{H})} \quad (\text{L.3})$$

The demagnetizing factor N_d is given by [1], as

$$\begin{aligned} \pi N_d = & \frac{b^2 - c^2}{2bc} \ln \left(\frac{\sqrt{a^2 + b^2 + c^2} - a}{\sqrt{a^2 + b^2 + c^2} + a} \right) + \frac{a^2 - c^2}{2ac} \ln \left(\frac{\sqrt{a^2 + b^2 + c^2} - b}{\sqrt{a^2 + b^2 + c^2} + b} \right) \\ & + \frac{b}{2c} \ln \left(\frac{\sqrt{a^2 + b^2} + a}{\sqrt{a^2 + b^2} - a} \right) + \frac{a}{2c} \ln \left(\frac{\sqrt{a^2 + b^2} + b}{\sqrt{a^2 + b^2} - b} \right) + \frac{c}{2a} \ln \left(\frac{\sqrt{c^2 + b^2} - b}{\sqrt{c^2 + b^2} + b} \right) \\ & + \frac{c}{2b} \ln \left(\frac{\sqrt{a^2 + c^2} - a}{\sqrt{a^2 + c^2} + a} \right) + 2 \arctan \left(\frac{ab}{c\sqrt{a^2 + b^2 + c^2}} \right) + \frac{a^3 + b^3 - 2c^3}{3abc} \\ & + \frac{a^2 + b^2 - 2c^2}{3abc} \sqrt{a^2 + b^2 + c^2} + \frac{c}{ab} \left(\sqrt{a^2 + c^2} + \sqrt{b^2 + c^2} \right) \\ & - \frac{(a^2 + b^2)^{3/2} + (b^2 + c^2)^{3/2} + (c^2 + a^2)^{3/2}}{3abc} \end{aligned}$$

where the dimensions of the magnet are $\pm a$, $\pm b$, and $\pm c$, with the axis along c being most-closely aligned with the magnetic field. Based on this result, [2] gives an apparent rod saturation flux

$$B'_s = \frac{\mu'_r B_s}{\mu_r^m} \quad (\text{L.4})$$

The theoretical calculation bounds of μ'_r and B'_s are upper limits themselves, e.g., they assume perfect manufacturing. Any manufacturing imperfections will result in less efficiency, which translates into a lower range of apparent permeability and therefore a lower range of saturation flux. This effect is discussed further in [2].

The energy dissipated per hysteresis cycle, D , by a damper is

$$D = V \oint \mathbf{H} d\mathbf{B} \quad (\text{L.5})$$

To find $\oint \mathbf{H} d\mathbf{B}$, which is the area within the permeable rod hysteresis loop, as well as establish the dynamics for the magnetic hysteresis torque, we use the hysteresis model based on [3], [4].

The major hysteresis loop is

$$\mathbf{B} = \frac{2B_S}{\pi} \tan^{-1}[k(\mathbf{H} \pm H_C)] \quad (\text{L.6})$$

where B_S is the saturation flux density, H_C is the coercive force, and k is a shaping factor

$$k = \frac{1}{H_C} \tan\left(\frac{\pi B_R}{2B_S}\right) \quad (\text{L.7})$$

where B_R is the remanence. The slope is

$$\left(\frac{d\mathbf{B}}{d\mathbf{H}}\right)_{ml} = \frac{2kB_S}{\pi} \cos^2\left(\frac{\pi \mathbf{B}}{2B_S}\right) \quad (\text{L.8})$$

The left and right boundary values for H are

$$H_L = \hat{H} - H_C \quad (\text{L.9})$$

$$G = \frac{H - H_L}{2H_C} \quad (\text{L.10})$$

$$H_R = \hat{H} + H_C \quad (\text{L.11})$$

$$G = \frac{H + H_R}{2H_C} \quad (\text{L.12})$$

where the term $\hat{H}$ is defined as follows:

$$\hat{H} = \frac{1}{k} \tan \left(\frac{\pi B}{2B_S} \right) \quad (\text{L.13})$$

The slope is

$$\frac{dB}{dH} = G^2 \left(\frac{dB}{dH} \right)_{ml} \quad (\text{L.14})$$

and

$$\frac{dB}{dt} = \frac{dB}{dH} \frac{dH}{dt} \quad (\text{L.15})$$

The material hysteresis loop is shown in Fig. L.1.

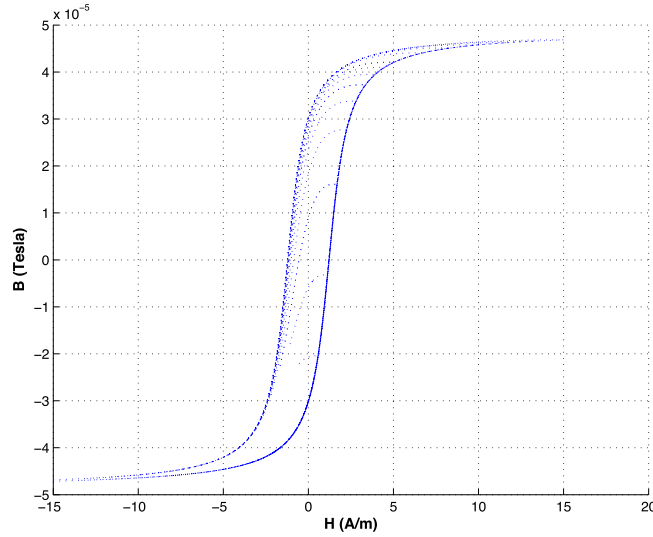


Figure L.1 Hysteresis for Permalloy C. The dotted lines are minor loops.

Integrating the major hysteresis loop over the maximum range of Earth's magnetic field at our altitude and inclination, which is ± 36.1 A/m, and inserting H_L and H_R into (L.6) to get the left-hand and right-hand curves, respectively, yields $2.44 \times 10^{-5} \text{ N/m}^2 \leq \oint H dB \leq 2.30 \times 10^{-4} \text{ N/m}^2$.

L.2. Energy-dissipation analysis

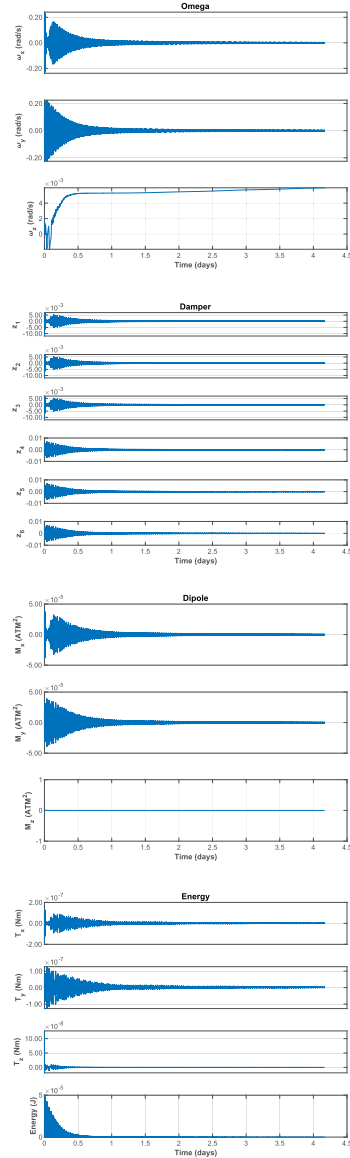
The average power dissipated is [5]

$$\dot{E} = -Df \quad (\text{L.16})$$

```

1 %% Constants
2 secInDay = 86400;
3 mu0 = 4e-7*pi;
4
5 %% User inputs
6 dateStart = [2023 6 1 0 0 0];
7 tDuration = 100*3600;
8 q0 = [1;0;0;0];
9 torqueD = [0;0;0]*1e-7;
10 tPulse = [1 2]*100000;
11 dT = 2;
12 omega0 = randn(3,1)*0.0001;
13 mRDamping = true;
14 d = RSHRigidBodyWithDamping;
15 d.dipole = [0;0;1];
16 [d.inertia, d.mass] = InertiaCubeSat('IU');
17 cDamp = 0.0001;
18
19 %% Start date
20 jDStart = Date2JD(dateStart);
21
22 %% Orbit
23 [el, jD0] = ISSOrbit;
24 el(6) = el(6) + (jDStart-jD0)*secInDay*2*pi /
    Period(el(1));
25 [r, v] = E12RV(el);
26
27 %% Time vector
28 n = ceil(tDuration/dT);
29 t = linspace(0, tDuration, n);
30 jD = jDStart + t/secInDay;
31
32 if (mRDamping)
33     d.dampingType = 0;
34     d.dampingData.Br = 0.004;
35     d.dampingData.Bs = 0.025;
36     d.dampingData.Hc = 12;
37     %% Damper rod unit vectors
38     d.dampingData.u = [0 0 0 1 1 1; 1 1 1 0 0 0;
39         0 0 0 0 0];
40     %% Dimensions are radius 1 mm by 95 mm
41     d.dampingData.v = pi*0.001^2*0.095*ones(1, size(
42         d.dampingData.u, 2));
43     d.dampingFun = @RSHHysteresisDamper;
44     uECI = QTForn(q0, d.dampingData.u);
45     [bl, blDot] = BDipole(r, jDStart, v);
46     hMag = Dot(uECI, bl) / mu0;
47     hMagDot = Dot(uECI, blDot) / mu0;
48     z = BFromHHysteresis(hMag, hMagDot, d.
49         dampingData);
50 else
51     %% constant damping
52     d.dampingType = 1;
53     z = zeros(6,1);
54     d.dampingData = cDamp;
55 end
56
57 %% Simulation loop
58 for k = 1:n
59     %% Plotting
60     [~, p] = RSHRigidBodyWithDamping(x, t(k), d);
61     omega = x(11:13);
62     energy = 0.5*omega'*d.inertia*omega;
63     xP(:, k) = [x(7:end); p.torqueDipole; p.
64         torqueDamper; energy];
65
66     %% Disturbance pulse for testing
67     if (t(k) > tPulse(1) && t(k) < tPulse(2))
68         d.torque = torqueD;
69     else
70         d.torque = [0;0;0];
71     end
72
73     %% Integrate
74     x = RK4(@RSHRigidBodyWithDamping, x, dT, t(k), d);
75 end
76
77 %% Plotting
78 yL = [d.states(:)'; 'M_x_u(ATM^2)'; 'M_y_u(ATM^2)';
79     'M_z_u(ATM^2)'; ...
80     'T_x_u(Nm)'; 'T_y_u(Nm)'; 'T_z_u(Nm)'; 'Energy_u(
81     J)'];
82
83 TimeHistory(t, xP(5:7,:), yL(11:13), 'Omega');
84 TimeHistory(t, xP(8:13,:), yL(14:19), 'Damper');
85 TimeHistory(t, xP(14:16,:), yL(20:22), 'Dipole');
86 TimeHistory(t, xP(17:20,:), yL(23:26), 'Energy');
87
88 Figui

```



Example L.1: Magnetic-hysteresis damping. The energy drops towards zero.

The damper torque is [3]

$$T = \frac{\nu b u^\times b_E}{\mu_0} \quad (\text{L.17})$$

where b is the damper axial flux density, ν is the volume of the damper, b_E is the Earth's magnetic field, u is the damper unit vector in the body frame, and μ_0 is the permeability of free space.

Example L.1 shows magnetic-hysteresis damping. The spacecraft has a 1-ATM² dipole along the z -axis. The script starts with a random angular rate. It takes several hours to damp the angular rates. The rates transverse to the magnetic torque rod go to zero.

References

- [1] A. Aharoni, Demagnetizing factors for rectangular ferromagnetic prisms, *Journal of Applied Physics* 83 (6) (March 1998).
- [2] F. Santoni, M. Zelli, Passive magnetic attitude stabilization of the UNISAT-4 microsatellite, *Acta Astronautica* 65 (5–6) (2009) 792–803.
- [3] R.R. Kumar, D.D. Mazanekand, M.L. Heck, Simulation and shuttle hitchhiker validation of passive satellite aerostabilization, *Journal of Spacecraft and Rockets* 32 (5) (September–October 1995).
- [4] T.W. Flatley, D.A. Henretty, A magnetic hysteresis model, in: K.R. Hartman (Ed.), *Flight Mechanics/Estimation Theory Symposium 1995*, May 1995, pp. 405–415.
- [5] V. Francois-Lavet, Study of passive and active attitude control systems for the OUFTI nanosatellite, Master's thesis, University of Liège, 2009–2010.